

CHANDANA SREE MALA

PHD RESEARCHER

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PROFILE SUMMARY

A third year PhD researcher and former data scientist with industrial experience of over two years. Currently specialising in Trustworthy AI with a focus on mitigating hallucinations in LLMs. Research interests include Retrieval Augmented Generation, Uncertainty Quantification and Explainability.

WORK EXPERIENCE

Doctor of Philosophy in A.I

University of Pisa and KDD Lab, Italy

2023 – Present

- Research is centred to building Trustworthy AI design pipelines particularly for mitigating hallucinations and making the models more faithful and robust.
- Published papers on Faithfulness Hallucinations in RAG systems, Explainability in RAG systems, Uncertainty estimation based Adaptive RAG, Explaining Behavioural shifts like sycophancy and hallucinations in LLMs and more.
- Interned at L3S Research Center, Leibniz University of Hannover, Germany for 4 months
- Interned at TU Delft, Netherlands for 3 months as a visiting researcher.
- Collaborated with Regione Toscana Italy, for a project based on Multimodal RAG and published a paper.
- Collaborated with other researchers to work on language models which focus on multiple perspectives and social aspects.

Data science analyst

2023

IQera Italia S.p.A, Rome, Italy

- Conducted descriptive and predictive statistical analysis on financial data for portfolio management.
- Analysed data of Non-performing loans (NPLs) and Unlikely to pay loans (UTPs) to forecast the recovery amounts and gross book values (GBVs) leveraging SOTA time series models ARIMA, SARIMAX, FB Prophet for credit risk management
- Developed M.L models for predicting the probabilities of credit risk and loan amount recovery.
- Collaborated with cross functional teams to enhance data accuracy and completeness, ensuring high-quality datasets for analysis

Data Scientist

2021 – 2023

Pragma etimos, Aprilia, Italy

- Built deep learning models for computer vision tasks, including image classification, object detection, and license plate recognition with YOLOv3,v4,v5 in Pytorch on COCO and PascalVOC datasets with mAP scores ranging between mAP@0.78 to mAP@0.92 for all versions of these models.
- Built a web application for Sentiment analysis of Italian sentences, leveraging Italian base BERT NLP model for entity recognitions, fine-tuned the model with trip advisor reviews data which is custom scraped and cleaned, the model evaluated with F1 score of 87%. Deployed the model with FastAPI Rest Framework.
- Utilized Power BI to develop KPIs and visualization of interactive finance dashboards and reports for effectively communicating insights to stakeholders.
- Developed RESTful APIs using Flask and Docker for deploying machine learning models.

Junior Data Analyst Intern
ARCA Dynamics, Rome, Italy

2019 – 2020

- Successfully worked on my thesis project to design and develop an experimental test bed for hardware in the loop simulation of star trackers by collaborating with this company.
- Analysed star sensors data and simulated the night sky images in the lab for magnitude, luminance, angular velocity calculations of the stars using pandas, numpy packages.

PROFESSIONAL SKILLS

AI/ML Technologies

- FoundationModels: Transformers, Large Language Models (LLAMA, MISTRAL, GPT-4, Qwen, Deepseek, BERT), Fine Tuning.
- Model Adaptation Techniques: Task Customization, LoRA, QLoRA,
- Retrieval-Augmented Generation: Information retrieval from web sources, external databases, building RAG pipelines

Tools & Frameworks

- Programming languages: Python, R, SQL, HTML
- Frameworks: TensorFlow, PyTorch, Power BI
- NLP Libraries: Hugging Face Transformers, spaCy, NLTK
- Data Handling: Pandas, NumPy, Sk-learn
- APIs and Deployment: FastAPI, Flask, Streamlit, Docker
- Version Control: Git, GitHub
- Explainability and Reliability: SHAP, LIME, LORE for model interpretability
- LLM Frameworks: LangChain, LLaMa Index, Lang Graph

EDUCATION

Master's in Aerospace Engineering

2017 – 2020

Sapienza University of Rome, Italy

Graduated with score 105/110 and awarded the Lazio Scholarship

Bachelor's in Mechanical Engineering

2013 – 2017

JNTUACEP, Pulivendula, India

Graduated with Grade of "First class with Distinction"

AWARDS AND CERTIFICATES

- Awarded 3 rd prize in XAI-Hackathon conducted by CNR Italy.
 - Awarded as Best Speaker and moderator in URF AI and Robotics International conference, Rome, Italy.
 - Participated in conferences HHAI 2024, Discovery Science 2024, FAIR 2024, ACDL 2024, Clic-it 2024.
 - Certified Indian National Cadet Corp(NCC) & National Social Service Scheme(NSS) Candidate
 - Certified in Python, Data science, ML, DL and NLP, PowerBI and Azure online courses.
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PUBLICATIONS

- **Mala, Chandana Sree**, Gizem Gezici, and Fosca Giannotti. "Disentangling Faithfulness Hallucinations in Retrieval-Augmented Generation: A Systematic Benchmark and Analysis". Machine Learning Journal by Springer.
- **Mala, Chandana Sree**, Gizem Gezici, Sezer Kutluk, and Fosca Giannotti. "FusionRAG-EX: An Interactive RAG Framework with Retrieval and Confidence-Aware Explanations" (2026). In ECML-PKDD 2026.
- **Mala, Chandana Sree**, Gizem Gezici, Sezer Kutluk, and Fosca Giannotti. "Trustworthy-by-Design: Building a Generative AI Chatbot for Italian Public Administration." (2025). In TRUST-AI, ECAI 2025.
- Gizem Gezici, **Mala Chandana Sree**, Fosca Giannotti, and Sezer Kutluk. "Responsible AI in Practice: Designing a Trustworthy GenAI Chatbot in Public Administration". Accepted as a Book Chapter and to appear in "Trustworthy Generative AI: Fundamentals, Applications, and Future Directions" open access book published by Springer 2025.
- **Mala, Chandana Sree**, Christian di Maio, Mattia Proietti, Gizem Gezici, Fosca Giannotti, Stefano Melacci, Alessandro Lenci, and Marco Gori. "Towards Building a Trustworthy RAG-Based Chatbot for the Italian Public Administration." In HHAi 2025, pp. 196-204. IOS Press, 2025.
- **Mala, Chandana Sree**, Simon Gottschalk, Gizem Gezici, Wolfgang Nejdl, Fosca Giannotti. "Can You Please Be Wrong? Analysing the Effects of Social Tone on Sycophancy and Hallucinations in LLMs"
- Muscato Benedetta, **Chandana Sree Mala**, Marta Marchiori Manerba, Gizem Gezici, and Fosca Giannotti. "An overview of recent approaches to enable diversity in large language models through aligning with human perspectives." In Proceedings of the 3rd Workshop on Perspectivist Approaches to NLP (NLPerspectives)@ LREC-COLING 2024, pp. 49-55. European Language Resources Association (ELRA), 2024.
- Cappuccio, Eleonora, Benedetta Muscato, Laura Pollacci, Marta Marchiori Manerba, Clara Punzi, **Chandana Mala**, Margherita Lalli, Gizem Gezici, Michela Natilli, and Fosca Giannotti. "Beyond Headlines: A Corpus of Femicides News Coverage in Italian Newspapers." In Proceedings of the 10th Italian Conference on Computational Linguistics (CLiC-it 2024), pp. 171-181. 2024.